“From Behavioral Segments to Persona” was demonstrated by using the dataset (/kaggle/input/lifestyle-dataset/genz_digital_lifestyle_tech_habits.csv). The dataset contained 500 rows representing real-world digital lifestyle of GenZ. The dataset was synthetic and created for research and analysis purposes. The variables used for this project were User_ID, Age, Gender, Daily_Screen_Time_hrs, Preferred_Device, Social_Media_Usage_hrs, Gaming_Activity, Tech_Savviness, Streaming_Subscriptions and Privacy_Awareness_Level. All analyses were carried out in R, using libraries such as FactoMineR, tidyverse, etc.

There were no missing values, no strong outliers, and no marked skewness. For analyses based on correlation or distance:

- Continuous variables were standardised to mean 0 and standard deviation 1.
- Categorical variables were converted to factors with appropriate ordering for the ordinal variables.

For methods that required a purely numeric matrix (Principal Component Analysis) or purely categorical matrix (Multiple Correspondence Analysis), the data were split into df_num (4 numeric variables) and df_cat (5 categorical/ordinal variables).

PCA was performed on the numeric subset, and I initially extracted as many principal components (PCs) as there were numeric variables. I retained the first two PCs as they captured a substantial proportion of the variance in the numeric variables.

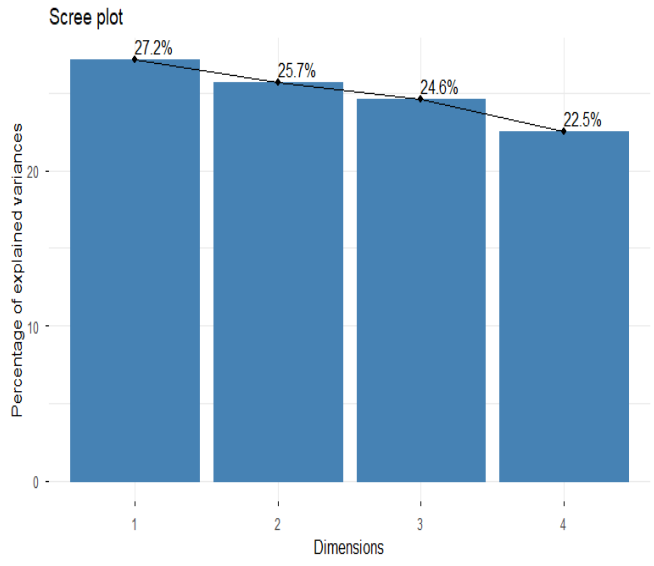


Figure (1) PCA scree plot with 4 components

MCA was conducted on the categorical subset, and the maximum possible number of MCA dimensions were requested. The eigenvalues declined gradually and each dimension accounted for around (6-10%) with no sharp elbow in the scree plot. For parsimony and interpretability, I decided to retain the first two MCA dimensions that captured the strongest underlying patterns.

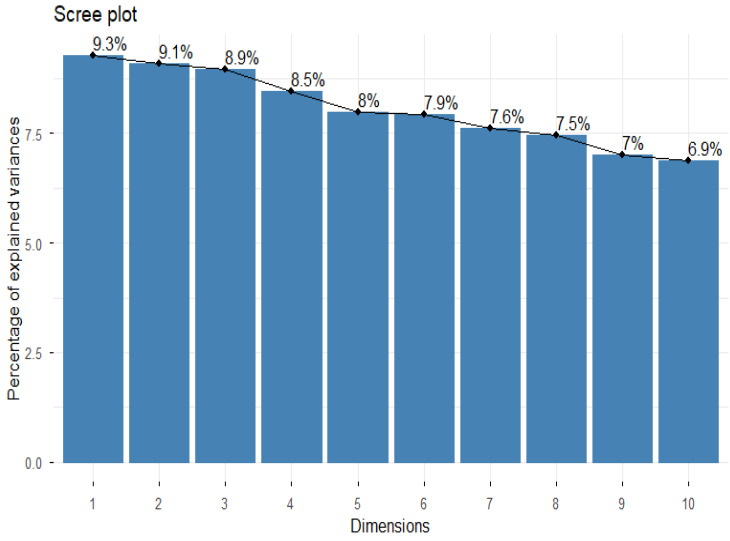


Figure (2) MCA scree plot with 10 components

To obtain a joint low-dimensional representation of the mixed data, I **combined the retained PCA and MCA scores**:

- For each individual, I concatenated:
    - PC1 and PC2 from the PCA (numeric part), and
    - Dim1 and Dim2 from the MCA (categorical part).
- This yielded a 4-dimensional feature vector per observation:
    - PC1, PC2, MC1, MC2 (where MC1 and MC2 are the first two MCA dimensions).
- The combined 4-dimensional matrix was then standardized, so that all components contributed comparably to the clustering distance.

Hierarchical agglomerative clustering was applied to the combined, scaled PCA+MCA scores. Specifically, Distance Measure (Euclidean distance) and Linkage method (Ward's minimum variance method) were applied. The resulting dendrogram summarized the nested grouping structure over all possible numbers of clusters (Figure – 3).

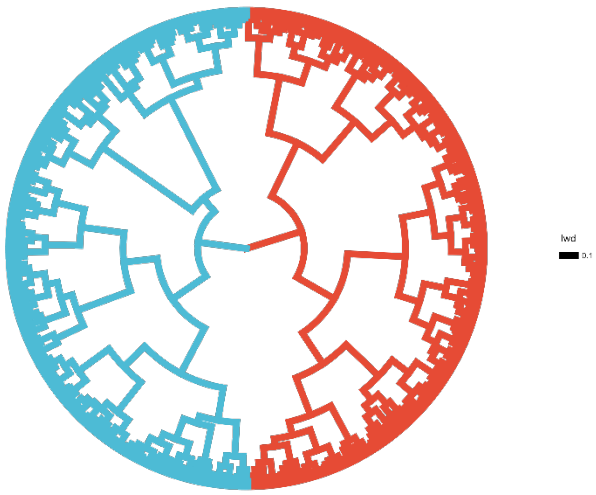


Figure (3) Hierarchical clustering on PCA+MCA scores (Dendrogram)

To select the number of clusters, I used the silhouette coefficient, which measures the cohesion and separation of clusters. Testing k from 2 to 7, mean silhouette values ranged from approximately 0.13 to 0.15. Despite the modest silhouette scores, two clusters (k = 2) were selected for subsequent analyses.

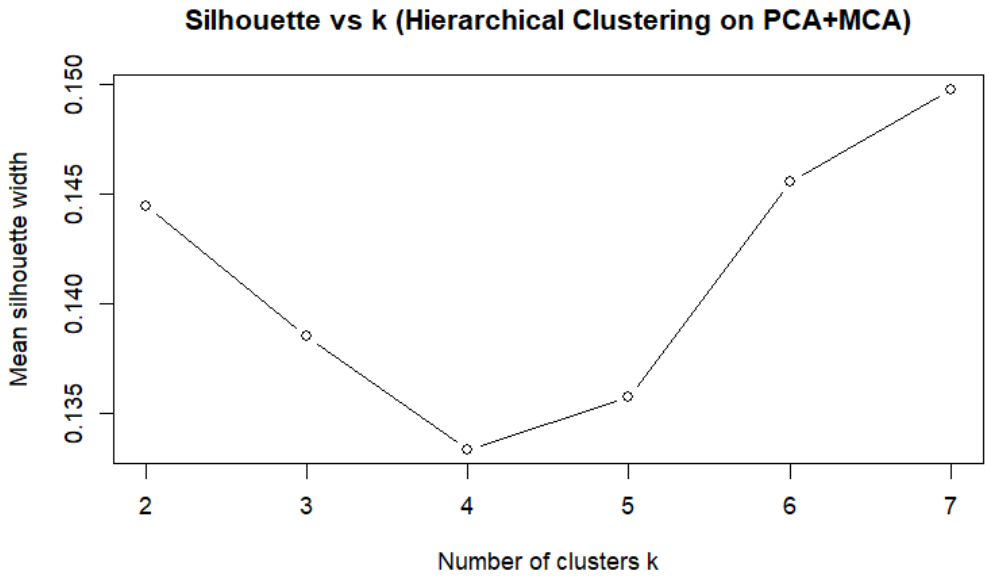


Figure (4) Graph showing Silhouette vs k

Interpretation of Dimensions

PCA on Numeric Variables

Dimension 1: 'High Digital Engagement'

- Streaming Subscription (0.750): This is the strongest positive link. Dimension 1 is mostly defined by people who engaged in more paid services.
- Social Media Usage (0.427) & Daily Screen Time (0.351): There is a moderate positive relationship with social media usage and daily screen time.
- Age (-0.468): It moves in the opposite direction.
- Interpretation: This dimension likely represents a user who is younger and heavily invested in paid streaming and social media.

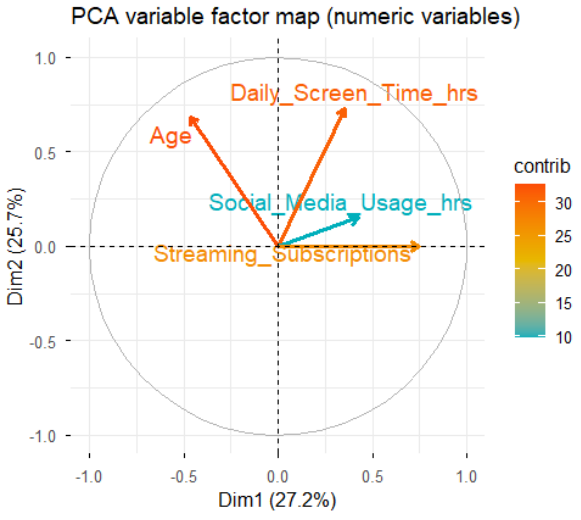


Figure (5) PCA variable factor map (Numeric variables)

Dimension 2: 'General Screen Exposure'

- **Daily Screen Time (0.731)** and **Age (0.686):** They have a strong positive influence here.
- **Social Media Usage (0.152):** It is weakly associated with the positive end.
- **Streaming Subscriptions (-0.001):** It has a very little negative influence on the relationship.
- **Interpretation**: This dimension captures "Time Spent" regardless of the specific platform, possibly older users who spend significant time on screen for purposes other than social media or streaming.

MCA on Categorical Variables

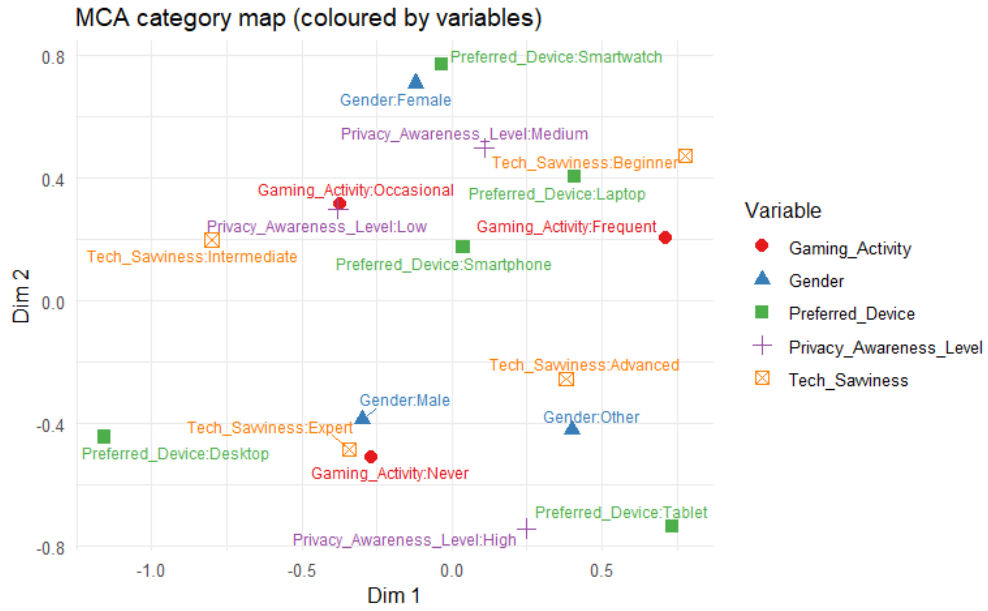


Figure (6) MCA category map (colored by variables)

Dimension 1: The positive side represents people who identify as 'Other' gender, mobile tech users (mostly tablets and laptops), frequent gamers, people with moderate to high privacy awareness level, people with beginner and advanced level tech skills, while males, females, desktop users, people who never play or play games, people with low privacy awareness level, people with intermediate and expert level tech savviness are on the negative side.

- **Gender:** "Other" gender type has a strong influence (0.400) on the positive side of Dimension (1), while "Male" and "Female" leans toward the negative side with (-0.296) and (-0.121).
- **Preferred Device:** Desktop users (-1.158) are at the extreme negative, and Smartwatch users (-0.032) sit closer to the center. Tablet users (0.733) and Laptop users (0.410) have a moderate positive relationship, whereas Smartphone users (0.038) are situated closer to the center.
- **Gaming Activity:** Dimension 1 seems to distinguish the frequent gamers (0.714) from the rest i.e., who never plays the game (-0.269) and who plays occasionally (-0.371).

- **Tech Savviness:** This axis is differentiating users based on whether they have moved past the initial learning curve with Beginner at (0.778) and Intermediate at (-0.799). Since Advanced (0.380) and Expert (-0.342) have lower absolute values, this dimension is less sensitive to distinguishing between the lower-to-middle skill tiers.
- **Privacy Awareness Level:** Low privacy awareness level (-0.380) is separated from medium and high levels at (0.107) and (0.249).

Dimension 2: Females, users with a focus on mobility (e.g. smartwatch, laptops), gamers, people with beginner to intermediate level tech skills, and people with low to medium privacy awareness level have a positive influence, whereas male and other gender type users, tablet and desktop users, non-gamers, people with advanced to expert level tech savviness, and people with high level privacy awareness have a negative influence.

- **Gender**: Females (0.709) are on the positive end and the other two (-0.387 and -0.422) are on the negative end.
- **Preferred Device**: Smartwatch users and Tablet users are on the far positive (0.769) and negative (-0.734) sides of Dimension (2). Meanwhile, Desktop and Laptop users are relatively closer to center with (-0.444) and (0.404) respectively; and the closet is the Smartphone users (0.174).
- **Gaming Activity**: Non-gamers have a strong negative coordinate (-0.508), creating a clear opposition against both 'Occasional' and 'Frequent' categories. This suggests that whatever characteristics define Dimension 2 are strongly *absent* or reversed for those who never game.
- **Tech Savviness**: The vertical axis follows a more linear progression of skill, separating Beginner (0.472) and Intermediate (0.197) from Advanced (-0.253) and Expert (-0.484).
- **Privacy Awareness Level**: The axis differentiates users with high privacy awareness (-0.742) from low (0.299) and medium (0.497) privacy awareness.

Interpretation of Clusters

Cluster 1: Desktop Traditionalist or Stationary Surfer

- Age: The older age group, with an average age around 24 years old
- Gender: The higher proportion of females
- Gaming & Tech Savviness: They are not big fans of gaming, and they lean more toward gaming occasionally. The highest proportion of 'Intermediate' level tech savviness.
- Behavior: With the higher proportion of low level of privacy concern, the cluster spends more time on screen, engaging more in social media and streaming services.
- Device: This group is heavily 'Desktop' focused.

Cluster 2: The young tech-forward users

- **Age:** The younger age group with an average age around 23 years old

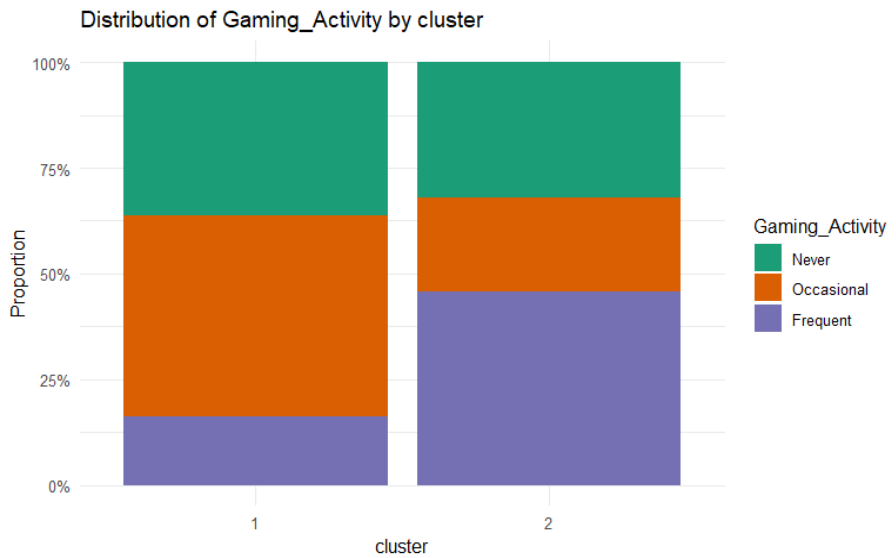


Figure (7) Distribution of Gaming Activity by Cluster

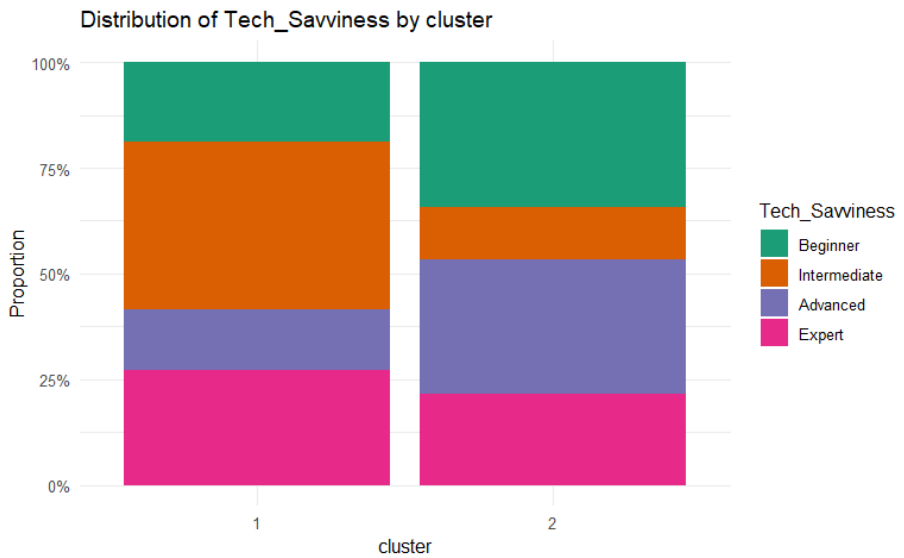


Figure (8) Distribution of Tech Savviness by Cluster

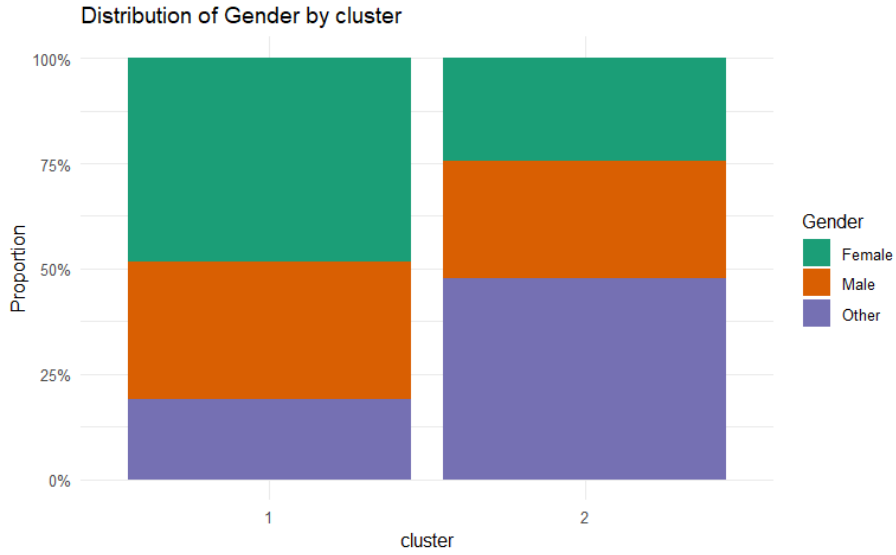


Figure (9) Distribution of Gender by Cluster

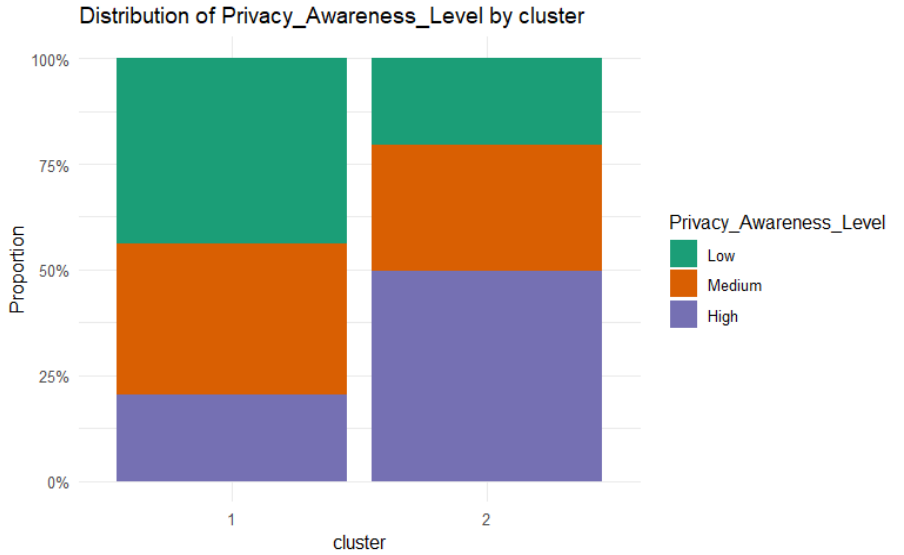


Figure (10) Distribution of Privacy Awareness Level by Cluster

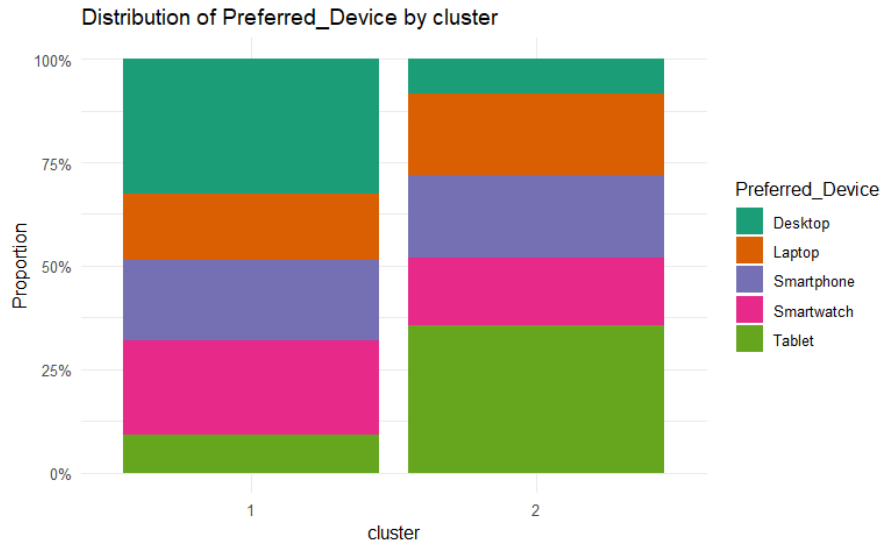


Figure (11) Distribution of Preferred Device by Cluster

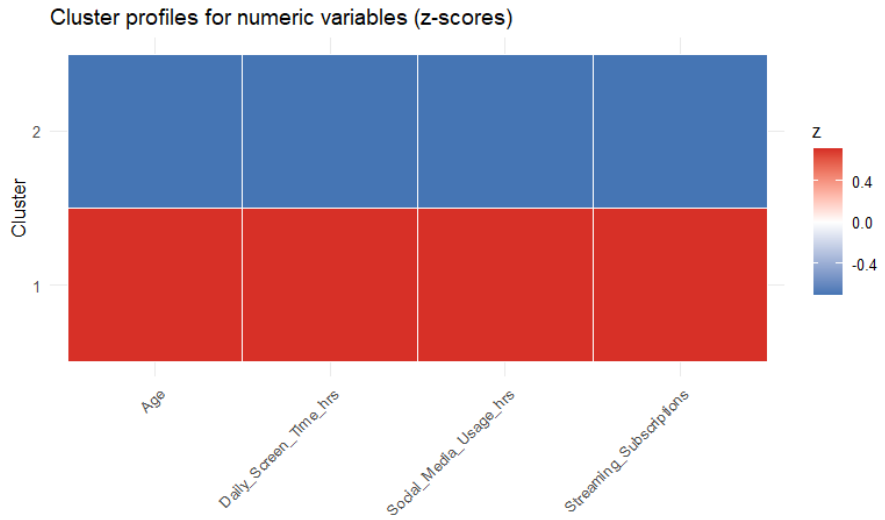


Figure (12) Cluster Profiles for Numeric Variables (z-scores)

- **Gender:** The higher proportion of 'Other' gender type
- **Gaming & Tech Savviness:** Higher overall gaming activity with 'frequent' gamers, and the higher proportions of beginner and advanced level tech astuteness representing a mix of hobbyists and tech skills focused individuals
- **Behavior:** With the higher proportion of 'high' privacy concern, the group engages less in social media and streaming services with reduced screen time.
- **Device:** The cluster focuses on portable devices (e.g., tablets).

Linking the Cluster's Profile to Dimension's coordinates

Cluster 1 Vs Dimensions

This cluster has a higher proportion of females, so females pull toward the top of Dim2. The older-aged group also moves toward the upper part of Dim2. While this cluster pulls toward the upper right space regarding daily screen time, they are situated near center on the right side of Dim1 regarding social media usage and streaming subscriptions. The higher proportion of occasional gamers moves toward the left side of Dim1. Having the largest proportion of 'intermediate' level tech astuteness, they pull toward the left side of Dim1. This cluster includes the

largest proportion of ‘low’ level privacy concern, so it is situated in the upper left quadrant. Given its ‘desktop’ preference, it is pulled to the lower left space.

Cluster 1 Persona: This seems to align with the profile of older individuals around 24 years old who have high daily screen time, engaging more in social media and paid streaming services. They may play a casual game once in a while. As they have a solid grasp of tech knowledge for daily life, they do not pay much attention to privacy. Females seem to represent this cluster, and they are generally desktop traditionalists or stationary surfers.

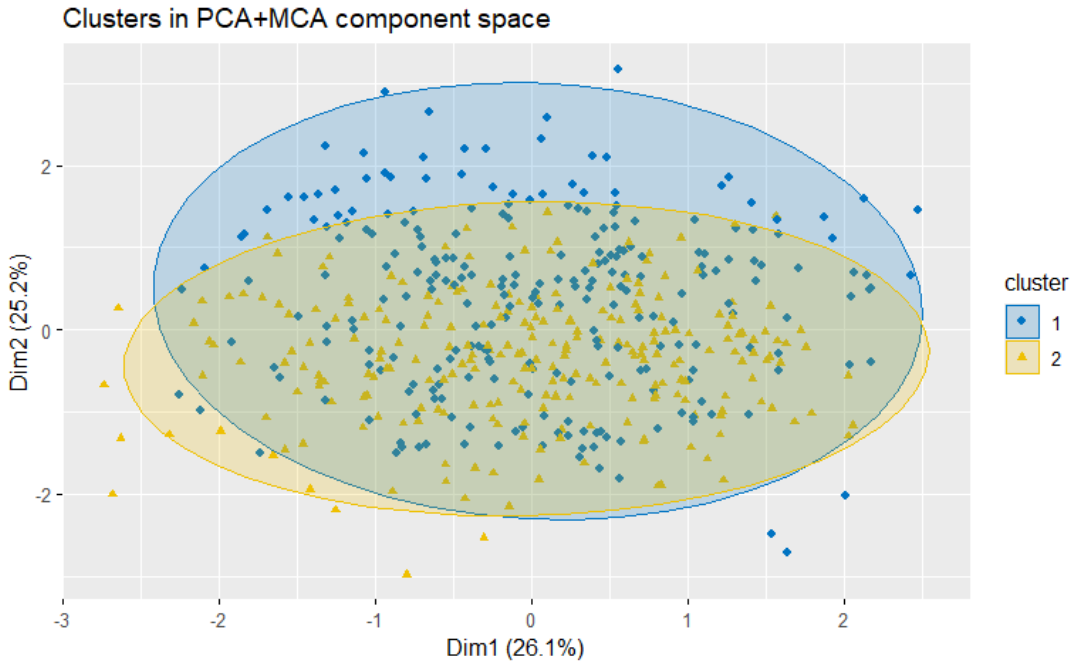


Figure (13) Clusters in PCA+MCA Component Space

Cluster 2 Vs Dimensions

This cluster includes the largest proportion of ‘other’ gender type and will be situated in the lower right space. The younger individuals (mean ~ 23) will be pushed to the left side of Dim1. While their daily screen time is on the right side (but closer to the center) of Dim1, social media usage is on the upper end of Dim 2 around the center, and the streaming subscription is almost on the center. Having a highest proportion of ‘frequent’ gamers, the group will be pushed toward the upper right space. The larger proportion of ‘beginner’ level tech savvy will sit on the upper right quadrant. With the largest proportion of ‘high’ level privacy awareness, this cluster will pull toward the lower right space. With a focus on the portable devices, the cluster is the top user of the tablets and will move to the right end of Dim1 and lower pole of Dim2.

Cluster 2 Persona: The younger ‘other’ gender dominant group with the average age of 22. Spending less time on screen, they engage less in social media and streaming services. They are active gamers with beginner level tech astuteness. With high level of privacy concern, portable devices, mainly tablets, are their favorite ones.